CO3 – AT2

Case Study based Assessment

SOURCE CODE:

```
#include <stdio.h>
void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int arr[], int low, int high)
{
    int pivot = arr[high]; // Last element as pivot
    int i = low - 1;
    for(int j = low; j < high; j++)
    {
        if(arr[j] > pivot) // Descending order for leaderboard
        {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return i + 1;
}
void quickSort(int arr[], int low, int high)
{
```

```c
    if(low < high)
    {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}
int main()
{
    int scores[] = {450, 120, 800, 350, 620, 980, 500};
    int n = sizeof(scores) / sizeof(scores[0]);
    printf("Original Leaderboard Scores:\n");
    for(int i = 0; i < n; i++)
        printf("%d ", scores[i]);
    quickSort(scores, 0, n - 1);
    printf("\n\nSorted Leaderboard (Highest Score First):\n");
    for(int i = 0; i < n; i++)
        printf("%d ", scores[i]);
    return 0;
}
```